

EXPERIMENT 3(a)

AIM:

To perform data visualization using a line plot to compare the sales of two products, Coke and Pepsi over different years.

CODE:

```
import matplotlib.pyplot as sales
```

```
Years = [2020, 2021, 2022, 2023, 2024]
```

```
Coke_Sales = [120, 135, 150, 165, 180]
```

```
Pepsi_Sales = [110, 125, 145, 155, 170]
```

```
sales.title("COKE Vs PEPSI")
```

```
sales.xlabel("Year")
```

```
sales.ylabel("Sales")
```

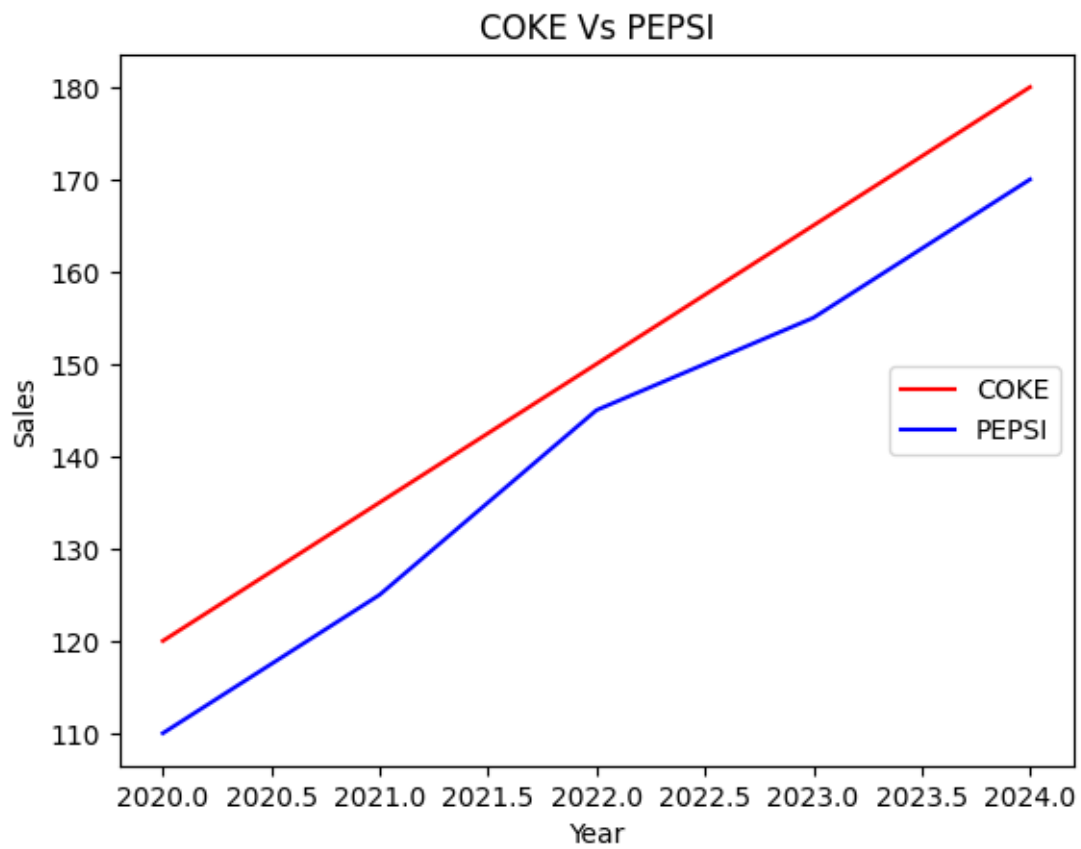
```
sales.plot(Years, Coke_Sales, color="red", label="COKE")
```

```
sales.plot(Years, Pepsi_Sales, color="blue", label="PEPSI")
```

```
sales.legend(loc="center right")
```

```
sales.show()
```

OUTPUT:



RESULT:

The sales data of Coke and Pepsi was successfully loaded and analysed. The line plot was generated successfully, and the missing values were present.

EXPERIMENT 3(b)

AIM:

To perform data visualization using a bar chart to compare the sales of different products.

CODE:

```
import matplotlib.pyplot as sales
```

```
import numpy as np
```

```
Products = ['Laptop', 'Mobile', 'Tablet', 'Watch', 'Headphone', 'Camera']
```

```
xaxis = np.arange(len(Products))
```

```
Sales = [85, 120, 65, 95, 110, 50]
```

```
sales.bar(Products, Sales)
```

```
sales.xticks(xaxis, Products, rotation=45)
```

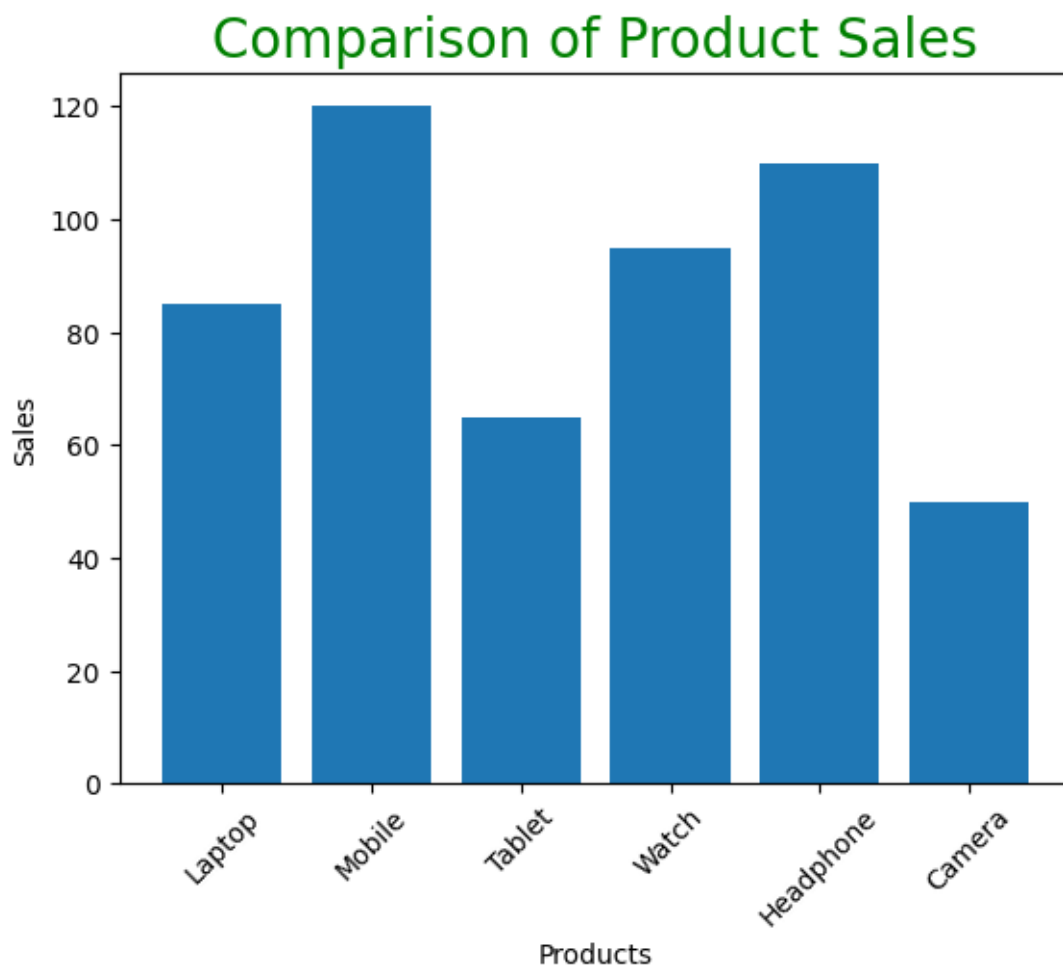
```
sales.xlabel("Products")
```

```
sales.ylabel("Sales")
```

```
sales.title("Comparison of Product Sales", fontsize=20, color="green")
```

```
sales.show()
```

OUTPUT:



RESULT:

The sales data of different products was successfully visualized using a bar chart, making it easy to compare the sales value of each product.

EXPERIMENT 3(c)

AIM:

To perform data visualization using a pie chart to represent the distribution of different expenses.

CODE:

```
import matplotlib.pyplot as expense

# Household expense data
labels = ['Food', 'Travel', 'Education', 'Shopping', 'Others']
Expenses = [3000, 2000, 4000, 2500, 1500]

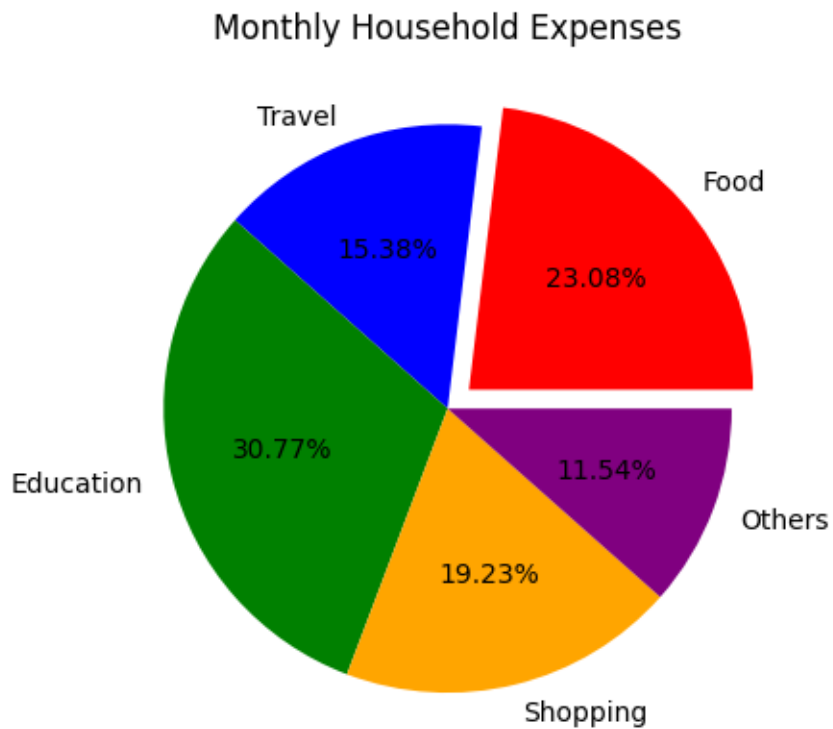
colors = ['red', 'blue', 'green', 'orange', 'purple']
explode = (0.1, 0, 0, 0, 0)

# Plotting the pie chart
expense.pie(Expenses, labels=labels, colors=colors,
            explode=explode, autopct='%0.2f%%')

expense.title('Monthly Household Expenses')

expense.show()
```

OUTPUT:



RESULT:

The distribution of monthly household expenses was successfully visualized using a pie chart, showing the percentage contribution of each expense category.